LAB 8

FUNCTION TEMPLATE

#1

```
template <typename any>
void swapp(any &a, any &b){
	any t = a;
	a = b;
	b = t;

}
int main(){
	int x=10,y=20;
	swapp(x,y);
	cout<<"Swapped values = "<<x<<"
"<<y;
	return 0;
}
```

```
Swapped values = 20 10
--------------------------------
```

Given were 10 and 20 (ints)

```
Swapped values = g h
----------------------------
```

Given were ‘h’ and ‘g’ (chars)

#2

```
template <typename any>
int minn(int n, any a[]){
	any min=a[0];
	for(int i=1;i<n;i++){
		if(a[i]<min)
			min=a[i];
	}
	return min;
}
int main(){
	int n; int arr[10];
	cout<<"Enter size of array: ";
	cin>>n;
	for(int i=0;i<n;i++){
		cout<<"Enter element: ";
		cin>>arr[i];
	}
	cout<<"Minimum Value =
"<<minn(n,arr);
	return 0;
}
```

```
Enter size of array: 5
Enter element: 3
Enter element: 4
Enter element: 10
Enter element: -2
Enter element: -3
Minimum Value = -3
```

#3

```
template <typename any>
void bubbles(int n, any a[]){
	for(int i=0;i<n;i++){
		for(int j=0;j<n-i-1;j++){
			if(a[j]>a[j+1]){
				any temp = a[j];
				a[j]=a[j+1];
				a[j+1]=temp;
			}
		}
	}
	cout<<"Sorted Array = "<<endl;
	for(int z=0;z<n;z++){
		cout<<a[z];
	}
}
int main(){
	int n; int arr[10];
	cout<<"Enter size of array: ";
	cin>>n;
	for(int i=0;i<n;i++){
		cout<<"Enter element: ";
		cin>>arr[i];
	}
	bubbles(n,arr);
	return 0;
}
```

```
Enter size of array: 5
Enter element: 2
Enter element: 1
Enter element: 3
Enter element: 4
Enter element: 5
Sorted Array =
12345
```

#4

```
template <typename any>
int linears(int n, any a[], any val){
	for(int i=0;i<n;i++){
		if(a[i]==val)
			return i;
	}
	return -1;
}
int main(){
	int n; char arr[10]; char val;
	cout<<"Enter size of array: ";
	cin>>n;
	for(int i=0;i<n;i++){
		cout<<"Enter element: ";
		cin>>arr[i];
	}
	cout<<"Element to search for?: ";
	cin>>val;
	int pos=linears(n,arr,val);
	if (pos!=-1)
		cout<<"Element found at index:
"<<pos;
	else
		cout<<"Element not found";
```

```
Enter size of array: 4
Enter element: a
Enter element: b
Enter element: c
Enter element: d
Element to search for?: g
Element not found
```

```
Enter size of array: 5
Enter element: 3
Enter element: 2
Enter element: 4
Enter element: 5
Enter element: 6
Element to search for?: 6
Element found at index: 4
```

```
        return 0;
}
```

#5

```
void process(){
        cout<<"This is the basic overloaded function"<<endl;
}

template <typename any>
void process(any val){
        cout<<"Overloaded function process with value ="<<val<<endl;
}

int main(){
        process();
        process(3.14);
        process('a');
        return 0;
}
```

```
This is the basic overloaded function
Overloaded function process with value =3.14
Overloaded function process with value =a
```

CLASS TEMPLATE

#1

```
#include <iostream>
using namespace std;

template <typename any>
class sstack{
        any arr[10];
        int top;
        public:
                sstack(){
                        top=-1;
                }
                void push(any val){
                        if(top>=9){
                                cout<<"Stack Overflow"<<endl;
                                return;
                        }
                        arr[++top]=val;
                }
                void pop() {
        if (top < 0) {
            cout << "Stack Underflow" << endl;
            return;
        }
        top--;
                }
                void display() {
```

```
c b a
b a
```

```
9.9 7.8 1.8
7.8 1.8
```

```
        if (top < 0) {
            cout << "Stack is empty" << endl;
            return;
        }
        for (int i = top; i >= 0; i--) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};
int main(){
    sstack <char> s;
    s.push('a');
    s.push('b');
    s.push('c');
    s.display();
    s.pop();
    s.display();
    return 0;
}
```

#2

```
#include <iostream>
using namespace std;

template <class T>
class Queue {
private:
    T arr[100];
    int front, rear;

public:
    Queue() {
        front = -1;
        rear = -1;
    }
    void enqueue(T value) {
        if (rear == 99) {
            cout << "Queue Overflow" << endl;
            return;
        }
        if (front == -1) front = 0;
        rear++;
        arr[rear] = value;
    }
    void dequeue() {
        if (front == -1 || front > rear) {
            cout << "Queue Underflow" << endl;
            return;
        }
        cout << "Deleted: " << arr[front] << endl;
        front++;
    }
```

```
10 20 30
Deleted: 10
20 30
```

```
    void display() {
        if (front == -1 || front > rear) {
            cout << "Queue is empty" << endl;
            return;
        }
        for (int i = front; i <= rear; i++) {
            cout << arr[i] << " ";
        }
        cout << endl;
    }
};

int main() {
    Queue<int> q1;
    q1.enqueue(10);
    q1.enqueue(20);
    q1.enqueue(30);
    q1.display();
    q1.dequeue();
    q1.display();
    return 0;
}
```

#3

```
#include <iostream>
using namespace std;

template <class T>
class Pair {
private:
    T a, b;
public:
    void setvalue(T x, T y) {
        a = x;
        b = y;
    }
    void display() {
        cout << a << " " << b << endl;
    }
};
int main() {
    Pair<int> p1;
    p1.setvalue(67, 420);
    p1.display();
    return 0;
}
```

```
67 420
```

#4

```
#include <iostream>
using namespace std;

template <class T>
class Calc {
private:
    T a, b;

public:
    void setvalue(T x, T y) {
        a = x;
        b = y;
    }
    void add() {
        cout << "Add: " << a + b << endl;
    }
    void sub() {
        cout << "Sub: " << a - b << endl;
    }
    void mul() {
        cout << "Mul: " << a * b << endl;
    }
    void div() {
        cout << "Div: " << a / b << endl;
    }
};
int main() {
    Calc<int> c1;
    c1.setvalue(100,20);
    c1.add();
    c1.sub();
    c1.mul();
    c1.div();
    return 0;
}
```

With ints 100,20

```
Add: 120
Sub: 80
Mul: 2000
Div: 5
```

With doubles 23.2,1

```
Add: 24.2
Sub: 22.2
Mul: 23.2
Div: 23.2
```

#5

```
#include <iostream>
using namespace std;

template <class T>
class Array {
private:
    T arr[100];
    int n;

public:
    void input() {
        cout << "Enter size: ";
        cin >> n;

        for (int i = 0; i < n; i++) {
```

```
Enter size: 4
a b c d
a b c d
```

```
        cin >> arr[i];
    }
}
void display() {
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}
};

int main() {
    Array<char> a1;
    a1.input();
    a1.display();
    return 0;
}
```